AUTOMATIC DETECTION OF TECTONIC SURFACE DEFORMATION IN INSAR UNWRAPPED MAPS USING DEEP LEARNING

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ABSTRACT

Interferometric Synthetic Aperture Radar (InSAR) is widely used to monitor earthquakes, volcanic activity, and landslides, but atmospheric noise often obscures true ground deformation signals. To solve this, we propose an improved Denoising Convolutional Neural Network (DnCNN) trained on synthetic InSAR unwrapped maps. The model utilizes residual learning to extract and remove noise components from interferograms. Application to the 2016 Mw 6.2 Tottori earthquake reveals a 10 cm line-of-sight (LOS) displacement, consistent with previous studies. Similarly, application to post-eruptive data from Mt. Ontake, Japan, identifies 17.6 cm of deflation, also in agreement with prior research. In both cases, reduced standard deviations further confirm the model's effectiveness for InSAR denoising.

***Index Terms*—** InSAR, deep learning, denoising

1. INTRODUCTION

Accurate surface deformation data is essential for studying crustal deformation mechanisms. Geodetic measurements are the only effective means to obtain precise surface displacement fields. Traditional methods, such as leveling, total stations, and GNSS, are time-consuming, labor-intensive, and limited in spatial and temporal coverage. In contrast, Interferometric Synthetic Aperture Radar (InSAR) offers high temporal and spatial resolution, wide coverage, and all-weather capability. It enables surface deformation measurements in remote or hazardous areas without ground control points [1], playing an irreplaceable role in geodesy. InSAR rapidly provides deformation data with accuracies from centimeters (coseismic) to millimeters per year (interseismic and postseismic), offering a quantitative basis for studying fault activity and crustal deformation throughout the seismic cycle.

However, challenges remain in interpreting InSAR results. Among them, atmospheric propagation delay, influenced by soil moisture and vegetation, is the most significant obstacle [2]. Due to tropospheric delay, current studies still rely on manual interpretation to identify low-amplitude tectonic deformation, such as slow-slip events, volcanic activity, post-seismic deformation, and hydrologically induced motion [3]. Therefore, efforts to extract tectonic signals from InSAR interferograms are both necessary and valuable.

Tectonic deformation and atmospheric noise show distinct spatial patterns in InSAR interferograms or unwrapped phase maps, enabling the use of computer vision-based deep learning models for denoising. Recent advances in deep learning have provided promising solutions for InSAR noise reduction. Previous studies have applied Convolutional Neural Networks (CNN) to extract seismic and volcanic deformation signals from noisy InSAR maps [2, 4], detect mining-related deformation [5], and used Recurrent Neural Networks (RNN) to identify landslide deformation [6]. However, due to the task-specific design of these architectures, such models are typically effective only for a single type of tectonic activity.

In this study, we propose an enhanced denoising convolutional neural network (DnCNN) to extract atmospheric noise from InSAR interferograms. The original DnCNN, designed for Gaussian noise suppression, is not directly applicable to InSAR due to its more complex noise characteristics [7]. To address this, we construct a tailored training dataset containing both signal and noise, and adjust patch and batch sizes to better reflect real-world InSAR conditions. Tests on synthetic datasets demonstrate the model's effectiveness in reducing noise in InSAR unwrapped maps.

2. METHODOLOGY

2.1. Architecture and Improvement of DnCNN

Fig. 1 illustrates the architecture and data flow of DnCNN during a single training or inference session. "Conv" denotes the convolutional layer for multi-scale feature extraction, "BN" the batch normalization layer for stabilizing training by maintaining consistent feature distributions, and "ReLU" the rectified linear unit introducing nonlinearity. For grayscale images, the standard DnCNN employs 64 convolutional filters of size 3×3 in the first layer to produce 64 feature maps. The network consists of 17 layers: an input layer, 15 intermediate layers for feature extraction, and an output layer. The first layer contains a Conv and ReLU block to extract initial features and map the input to feature space. Each intermediate layer includes Conv, BN, and ReLU block

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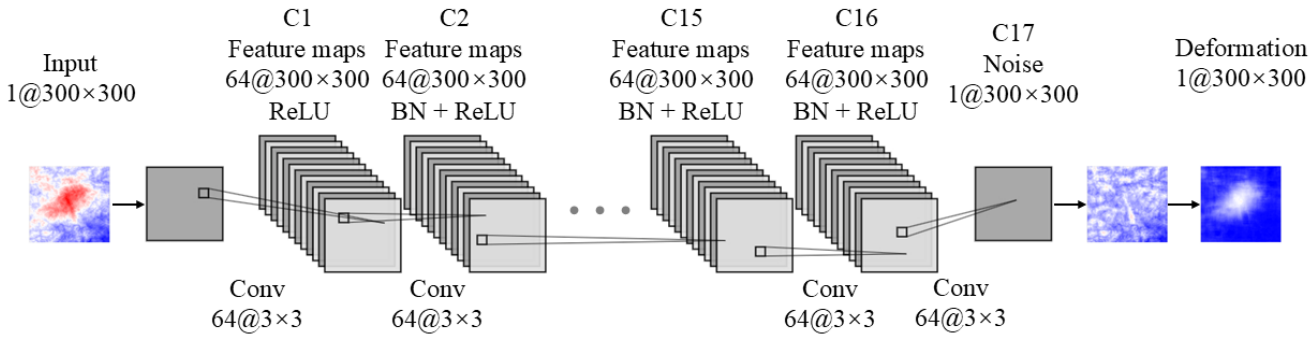

Fig. 1. Structure of DnCNN architecture.

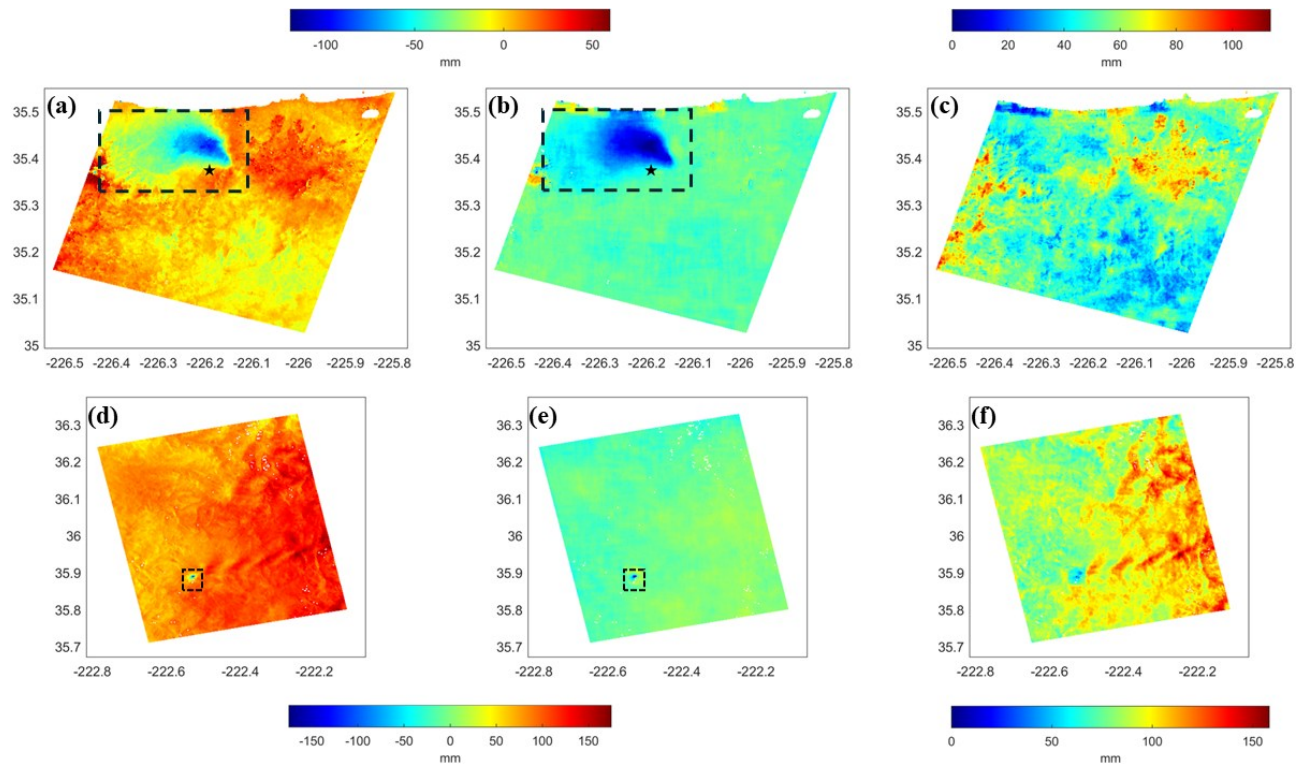

Fig. 2. Application of the proposed method to real InSAR unwrapped maps. (a) and (b) show the noisy and denoised unwrapped maps for the 2016 Mw 6.2 Tottori earthquake, respectively. (c) shows the predicted noise corresponding to (a). (d) and (e) show the noisy and denoised unwrapped maps for Mount Ontake, Japan. (f) shows the predicted noise corresponding to (d). The black rectangle indicates the verified deformation area, and the black pentagram marks the earthquake epicenter.

for progressive feature refinement. The final layer uses a Conv block to aggregate the 64 feature maps into a single noise map.

DnCNN is based on residual learning [8]. Let y denote the noisy image, x the clean image, and n the additive noise, such that: $y = x + n$. Instead of directly predicting the clean image x, DnCNN is trained to estimate the residual noise n. Accordingly, the loss function is defined as:

$$l(\theta) = \frac{1}{2n}\sum_{i=1}^{n}\|R(y,\theta) - (y - x)\|^2 \qquad (1)$$

Here, n denotes the batch size, i.e., the number of images per training iteration. $R(y,\theta)$ represents the noise predicted by the model, with θ denotes its parameters.

The original DnCNN was designed to remove high-frequency Gaussian noise from natural images. However, InSAR interferograms exhibit non-Gaussian, non-uniform, and spatially varying noise, making direct application of DnCNN suboptimal [9]. Moreover, real InSAR data often contain millions of pixels over areas spanning tens to hundreds of kilometers, which is incompatible with the original 50×50-pixel training patches. Therefore, in this study, we modify the network architecture to better accommodate the characteristics of InSAR unwrapped phase maps.

To balance performance and training constraints, we set the patch size to 300×300 pixels—covering about 50 km²—to capture fine-scale features while preserving spatial realism.

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A batch size of 16 is used to ensure training stability and computational efficiency.

2.2. Synthetic InSAR Unwrapped Maps

InSAR processing computes the phase difference between two Single-Look Complex (SLC) images. The resulting interferogram reflects the wrapped phase—relative phase modulo half the radar wavelength—ranging from -2π to 2π. Its periodic nature makes noise in the wrapped phase harder to distinguish from deformation signals, complicating training with models designed for unwrapped phases. Therefore, our study uses synthetic unwrapped InSAR maps, which directly represent ground deformation and enable more effective noise pattern learning. To design a generalized model, we assume an unwrapped InSAR map comprises four components: (1) tectonic deformation, (2) tropospheric hydrostatic delay, (3) tropospheric turbulent delay, and (4) linear orbital ramp.

For the tectonic deformation component, surface displacements were simulated using the Okada model or Mogi source [10, 11] with randomly sampled parameters, including depth, dip, rake, strike, width, length (Okada), and volume change (Mogi). Tropospheric hydrostatic noise was simulated by generating a Shuttle Radar Topography Mission (SRTM) digital elevation model (DEM) [12] over a 50 km² area (54°E–120°E, 26°N–37°N), represented as a 2D Gaussian surface with 166 m pixel spacing. Hourly global Zenith Total Delay (ZTD) data from the HGTP model were integrated with the DEM using the Iterative Tropospheric Decomposition (ITD) method [13, 14, 15] to produce the hydrostatic delay field. The tropospheric turbulent delay was modeled as spatially correlated noise with covariance between locations i and j given by an exponential function [16], expressed as:

$$C_d^{ij} = e^{-d^{ij}/L_c} \tag{2}$$

$$C = L \times L^T \tag{3}$$

$$N_t = L \times N_n \tag{4}$$

Here, C_d^{ij} denotes the covariance between locations i and j; d^{ij} is their Euclidean distance; L_c controls the spatial decay of covariance; C is the full covariance matrix with Cholesky decomposition L; N_n is a standard Gaussian noise vector, and N_t is the resulting turbulent delay. Long-wavelength orbital ramp errors were simulated by a bilinear ramp $\varphi = ax + by + c$ with random parameters. Each noise component was normalized individually, then summed and normalized again. To control noise level, a dynamic noise index $\beta \sim \mu(0.6, 1.2)$ adjusts noise relative to deformation. The final synthetic unwrapped map is: $\Phi = \beta\Phi_{noise} + \Phi_{def}$ where Φ is the synthetic unwrapped phase, Φ_{noise} the combined normalized noise, and Φ_{def} the simulated tectonic deformation.

3. RESULTS AND DISCUSSION

3.1. Data Generation and Training

This chapter evaluates the model's performance and its application to real InSAR unwrapped maps. Using our synthetic data generation method, we produced 60,000 noisy-clean map pairs, split into 50,000 for training, 9,000 for testing, and 1,000 for validation. To prevent overfitting and improve generalization, we employed a dynamic shuffling strategy [17], randomly pairing noisy and clean maps each epoch rather than pre-generating the entire dataset. With 50,000 clean and 50,000 noisy maps, up to 2.5×10^9 unique combinations exist. Each epoch samples 50,000 pairs, ensuring statistically different mini-batches across epochs, enhancing robustness. The model was trained for 180 epochs on an NVIDIA RTX 3090 GPU, with batch size 16 and initial learning rate 1×10^{-3}, requiring 60 hours of GPU time. Performance was then tested on the reserved test set.

3.2. Validation and Performance Test

$$SSIM = \frac{2\mu_x\mu_{y+C_1}}{\mu_x^2\mu_y^2C_1} \times \frac{2\sigma_{xy}+C_2}{\sigma_x^2+\sigma_y^2+C_2} \tag{5}$$

$$PSNR = 10 \times log\left(\frac{MAX_I^2}{MSE}\right) \tag{6}$$

$$SNR = \frac{RMS^2(Signals)}{RMS^2(noise)} \tag{7}$$

Given the distinct spatial characteristics of descending and ascending InSAR interferograms, separate deep learning models were trained for each track. After 180 epochs, the best models were selected based on validation datasets. Denoising performance was evaluated using Structural Similarity Index (SSIM) and Peak Signal-to-Noise Ratio (PSNR), defined in Equations (5) and (6), where μ_x, μ_y are image means, σ_x, σ_y standard deviations, c_1, c_2 stabilizing constants, Mse is the mean squared error, and Max is the maximum pixel value. Using Pareto frontier analysis [18], the best models were the 24th epoch for ascending data (mean SSIM 0.80, PSNR 20.65 dB) and the 103rd epoch for descending data (mean SSIM 0.79, PSNR 20.96 dB). Testing was conducted on separate data. To further assess denoising, Signal-to-Noise Ratio (SNR), defined as in Equation (7), was compared with SSIM. Defining a failure threshold at SSIM < 0.7 when performance test, The ascending model performs well when SNR > 0.07, and the descending model when SNR > 0.06. These results confirm the model's effectiveness even when noise dominates InSAR maps.

3.3. Application and discussion

After validation and testing, we applied our model to two real cases using ALOS-2/PALSAR-2 data. The model's denoising performance was assessed via the standard deviation of unwrapped phase maps and compared with previous studies.

In the first case, we evaluated the model on coseismic deformation within noisy unwrapped maps from the 2016

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Mw 6.2 Tottori earthquake. Descending ALOS-2/PALSAR-2 data (tracks 27-2870, 12 Dec 2014–23 Oct 2016) were processed using GMTSAR [19] and unwrapped via SNAPHU [20]. The unwrapped maps were downsampled to 166 m pixel spacing to match synthetic data resolution. As shown in Fig. 2 (a) and (b), the standard deviation decreased from 18.63 mm to 14.40 mm after denoising. The coseismic LOS deformation reduced from ~16 cm to 10 cm, consistent with the 120 mm reported by the Geospatial Information Authority of Japan (GSI).

In the second case, we tested the model's denoising in a mountainous volcanic region using ascending track 126-710 (3 Oct 2014–12 Jun 2015). As shown in Fig. 2 (d) and (e), Following the same processing, the standard deviation dropped from 21.19 mm to 9.60 mm after denoising, and the maximum post-eruptive LOS deformation decreased from ~33 cm to 17 cm. These results agree with the 21 cm/yr LOS rate reported by Narita et al. (2018), who corrected atmospheric errors by removing elevation-correlated components [21].

4. CONCLUSION

This article presents the use of a residual learning network, DnCNN, to reduce noise in InSAR unwrapped maps. After 180 training epochs on synthetic datasets, the model was applied to two real cases: coseismic deformation from the 2016 Mw 6.2 Tottori earthquake and post-eruptive deflation at Mt. Ontake, Japan. In both cases, noise was effectively reduced. Future work will apply the model to more InSAR interferograms with diverse tectonic settings to further validate its robustness.

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